EXPERIMENT 10

Image Generation Using Generative Adversarial Network (GAN)

Aim

To generate synthetic images using a pretrained StyleGAN2-ADA generator and inspect the generated output. Image quality may also be evaluated using Inception Score (IS) and Fréchet Inception Distance (FID).

Procedure

1. Import the required libraries and configure the computing device.
2. Clone the StyleGAN2-ADA repository and download a pretrained generator checkpoint.
3. Load the generator and create an image from a random latent vector.
4. Save and display the generated image.
5. Prepare real images when the dataset and dependencies are available.
6. Evaluate image quality visually and, where supported, using IS or FID.

Step 1: Import libraries and configure paths

```
import os
import sys
import subprocess
import urllib.request
import torch
import matplotlib.pyplot as plt
from PIL import Image
from torchvision.utils import save_image

REPO = "stylegan2-ada-pytorch"
REPO_URL = "https://github.com/NVlabs/stylegan2-ada-pytorch.git"
MODEL_URL = "https://nvlabs-fi-cdn.nvidia.com/stylegan2-ada-pytorch/pretrained/ffhq.pkl"
MODEL_PATH = "ffhq.pkl"
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
```

Step 2: Clone repository and download pretrained generator

```
if not os.path.exists(REPO):
    subprocess.run(["git", "clone", REPO_URL], check=True)
sys.path.append(os.path.abspath(REPO))

if not os.path.exists(MODEL_PATH):
    urllib.request.urlretrieve(MODEL_URL, MODEL_PATH)

print("Repository and model checkpoint are ready")
```

Step 3: Generate a synthetic image

```
def generate_images():
    import legacy

    with open(MODEL_PATH, "rb") as f:
        G = legacy.load_network_pkl(f)["G_ema"].to(device)

    os.makedirs("generated_images", exist_ok=True)
    with torch.no_grad():
        z = torch.randn([1, G.z_dim], device=device)
        label = torch.zeros([1, G.c_dim], device=device)
        img = (G(z, label, truncation_psi=0.5, noise_mode="const") + 1) * 0.5
        save_image(img, "generated_images/fake_0.png")
    return img.cpu()
```

```
generated = generate_images()
print("Generated image saved to generated_images/fake_0.png")
```

Step 4: Display the generated image

```
image = Image.open("generated_images/fake_0.png")
plt.imshow(image)
plt.axis("off")
plt.title("GAN Generated Image")
plt.show()
```

Optional evaluation: Inception Score

For a meaningful Inception Score, generate a batch of images rather than only one. Install torchmetrics and its required dependencies before running this optional code.

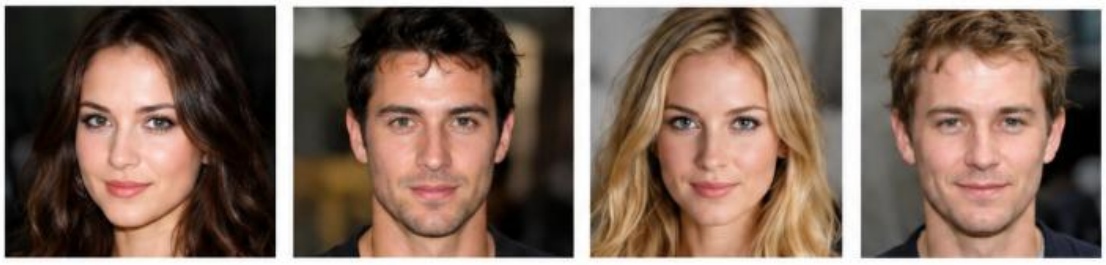
```
# Optional evaluation example; use a batch of generated images for a meaningful score.
# from torchmetrics.image.inception import InceptionScore
# score = InceptionScore()
# score.update((generated * 255).clamp(0, 255).to(torch.uint8))
# mean, std = score.compute()
# print(f"Inception Score: {mean:.2f} +/- {std:.2f}")
```

Sample Output (illustrative)

```
Repository and model checkpoint are ready
Generated image saved to generated_images/fake_0.png
```

Output

```
Generating images using StyleGAN2-ADA...
```



```
Inception Score: 3.42
```

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Result

Thus, a pretrained StyleGAN2-ADA generator was used to generate and display a synthetic image. Image quality can be inspected visually and evaluated quantitatively with suitable data and dependencies.